

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

*I **Aaron Samuel S** (192411022)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with	Incorrect or irrelevant answer

				noticeable errors	
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Responsive Web Design (RWD).

Responsive web Design is a web development approach that ensures a website automatically adjusts its layout and content according to the screen size and resolution of diff. devices such as mobiles, tablets & desktops

★ Importance :-

- Provides a consistent user experience across all devices
- Improves website accessibility and readability
- Reduces the need for separate mobile and desktop websites
- Improves SEO, as search engines prefer mobile friendly websites
- Increases user engagement and reduces bounce rates

★ Key CSS Tech. for Responsiveness.

1. Flexible layouts - Use % or viewport units (vw, vh) instead of fixed pixels

2. Flexible Images - Apply :

```
img {  
    max-width: 100%;  
    height: auto;
```

}

3) Media Queries - Apply diff. styles for diff screen sizes

@ media (max-width : 768px) {

• container {

flex-direction : column;

}

}

4) Responsive Typography

Use relative units like em, rem or vw

★ Result:-

- Desktop → Items appear side by side
- Mobile → Items appear one below another

★ Improvements:-

- Optimize images for faster loading
- Use touch friendly buttons and menus
- Apply lazy loading for images
- Maintain sufficient spacing and readable fonts
- Test the website on multiple devices and browsers.

2. Client Side Validation

Client Side validation checks user input in the browser before sending data to the server

★ Purpose:-

- Prevents invalid data submission
- Improves user experience through instant feedback
- Reduces unnecessary server requests
- Saves server processing time.

★ JS Event Handling:-

It validates:-
• onsubmit ; • oninput ; • onblur ; • onchange.

eg:-

```
<form onsubmit = "return validateForm()">
```

Email

```
<input type = "text" id = "email">
```

```
<input type = "submit">
```

```
</form>
```

```
<script>
```

```
function validateForm () {
```

```
    let email = document.getElementById("email").value
```

```
    let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

```
    if (!pattern.test(email)) {
```

```
        alert ("invalid email");
```

```
        return false;
```

```
    }  
    return true
```

```
</script>
```

★ Role of Regular Expression (Regex).

Regex is used to match input with a required pattern

* Email validation

`/^[^\s@]+@[^\s@]+\.[^\s@]+$/`

* Phone no. validation.

`/^[6-9]\d{9}$/`

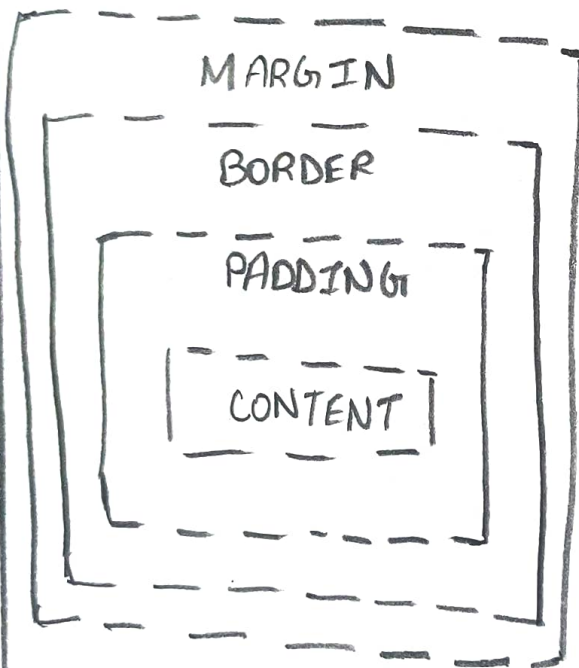
★ Improvements:-

- Display error messages beside input fields
- Validate while typing
- Use HTML5 validation attributes
- Always perform server-side validation for security
- Sanitize user input to prevent attacks.

3. CSS Box Model:-

Every HTML element consists of 4 parts (ie) within a rectangular box.

1. Content - Actual text or image
2. Padding - Space b/w content and border
3. Border - Surrounds the padding
4. Margin - Space outside the border.



★ CSS Positioning Tech.

- Static:- Default position
- Relative:- Moves relative to its position
- Absolute:- Positioned relative to nearest positioned parent
- Fixed:- fixed to the browser window
- Sticky:- Acts as relative until scrolling reaches a limit.

★ Causes of Layout Breaking:-

- Fixed width elements
- Incorrect positioning
- Missing media queries

★ Using Media Queries:-

@ media (max-width: 768px) {

• container {

display: block;

}

img {

width: 100%;

}

}

★ Best practices:-

- Use flexbox or grid
- Use relative units instead of fixed pixels
- Optimize images
- Apply media queries

4. DOM - Doc. object Model.

DOM is a programming interface that represents an HTML doc. as a tree of objects. JS uses the DOM to access and modify webpage elements dynamically without reloading the page.

★ Role of DOM:-

- Access HTML elements
- Modify text and content
- Change CSS styles

★ DOM eg:-

```
<h2 id="msg"> Welcome </h2>
```

```
<button onclick="changeText()">
```

click Here

```
</button>
```

```
<script>
```

```
function changeText () {
```

```
document.getElementById("msg").innerHTML = "welcome to JS";
```

```
}
```

```
</script>
```

★ Output:-

Initially: Welcome

After clicking the button: Welcome to JS

★ Common DOM Method:-

- getElementById()
- createElement()
- appendChild()
- removeChild()
- querySelector()

★ Improvements:-

- Min. unnecessary DOM updates
- Use event delegation for better perf.
- Cache frequently used DOM elements
- Update only required parts of the page.